STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 5: Work, Energy and Power

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Equivalent correct methods should receive credit.

Q1 1 mark

When force is perpendicular to displacement.

Suggested allocation by section: 1-mark questions earn one mark for the correct fact, formula or unit; 2-mark questions normally award one mark for the principle/formula and one for the explanation or result; 3-mark questions award credit for the relation, working/reasoning and result with unit; 4-mark questions award credit for the principle, equation, working and final answer/interpretation. Accept equivalent correct methods.

Q2 1 mark

mv^2.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q3 1 mark

Watt.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q4 1 mark

Net work equals change in kinetic energy.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q5 1 mark

kx^2.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q6 1 mark

Momentum and kinetic energy.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q7 1 mark

mgh.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q8 1 mark

Zero.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q9 2 marks

40 J.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q10 2 marks

24 J.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q11 2 marks

2000 J.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q12 2 marks

0 J.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q13 2 marks

1 J.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q14 2 marks

1 m/s.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q15 3 marks

32 J.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q16 3 marks

sqrt60~7.75 m/s.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q17 3 marks

18 J.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q18 3 marks

-200 kJ; 4000 N.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q19 4 marks

10 m/s; sqrt80~8.94 m/s.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.

Q20 4 marks

2 m/s; 24 J lost.

Suggested credit: award marks for the correct principle, working and result as applicable; accept equivalent correct methods.